

Gated Progressive Formalization — Operational Upgrade

Sharper gates, claim-type pre-classification, scored checklists, exemplars, calibration notes, failure-mode logging, and ready-to-use templates

1. Claim-Type Pre-Classification (New Required Step)

Before scoring gates, every major claim or candidate layer element is flagged as primarily **Descriptive** (factual, causal, or structural) or primarily **Normative/Strategic** (value, advocacy, prescription, or framing recommendation). Mixed claims are split or dual-flagged. This annotation is recorded and used when interpreting Cons and Amb. Descriptive claims are held to tighter factual constraint; normative/strategic claims are expected to carry higher residual Amb unless tightly linked to constrained descriptive premises.

2. Sharper, Measurable Amb (Ambiguity / Under-determination)

Amb is now scored by explicit free-parameter count and severity rather than pure impression.

Counting rule

A free parameter is any of: (a) an unspecified functional form or quantitative weight that materially affects predictions; (b) an alternative reading of a key term or causal link that remains equally compatible with all current anchors and admitted material; (c) a scope or boundary condition left open that changes the claim's implications. Count each distinct free parameter once.

Severity weighting (simple)

- High-severity: changes the core prediction or classification of the claim (weight 2).
- Medium-severity: changes secondary predictions or scope (weight 1).
- Low-severity: stylistic or peripheral (weight 0.5).

Score interpretation (guidance)

- Weighted free-parameter sum $\leq 2 \rightarrow$ Amb generally low enough for admission if other gates pass.
- Sum 3–5 $\rightarrow$ provisional / borderline; require explicit justification or further search.
- Sum $\geq 6 \rightarrow$ Amb high; block expansion of the formal layer.

Borderline examples

- "G is a three-component innate vector" after developmental anchors $\rightarrow$ low Amb on contents of G (previously high).
- "DSA is the heart of the Democratic Party" after urban-primary wins only $\rightarrow$ still high Amb (representativeness and national scope remain free).
- "The database rollout was unusually broad relative to the taxable set and paired with celebratory targeting" $\rightarrow$ low-to-medium Amb (factually constrained).

3. Sharper, Measurable Prod (Productivity)

A consequence counts as productive only if it is (1) new relative to the anchors and prior layers, (2) non-trivial (not a near-paraphrase), and (3) checkable in principle against observation, data, or further constrained inference.

Scoring rule

- 0 productive consequences $\rightarrow$ fails Prod.
- 1 clear productive consequence $\rightarrow$ minimum pass.
- 2+ independent productive consequences $\rightarrow$ strong Prod.

Borderline examples

- "If relational input is removed, C should stall near G" $\rightarrow$ productive (checkable developmental prediction).
- "Socialism is dangerous" (bare) $\rightarrow$ not productive (too vague to check).
- "DSA-aligned candidates won specific named 2026 NYC primaries" $\rightarrow$ productive once documented.

4. Scored Gap-Ranking Checklist

For each identified free parameter / gap, score 0–2 on each dimension and sum:

- **Impact:** How many other free parameters or core predictions would a constraint affect? (0=few, 1=several, 2=many/core)
- **Anchor connection:** How directly does the gap touch existing anchors? (0=remote, 1=moderate, 2=direct)
- **Measurability:** How feasible is finding constraining material? (0=very hard, 1=moderate, 2=relatively feasible)

Attack highest-sum gaps first. Record scores. This does not prove "most important"; it reduces pure arbitrariness.

5. Classical Sound-Argument Exemplars (Initial Set Guidance)

Begin domain-specific exemplar sets with arguments that historically closed important free parameters. Example starter categories: (a) developmental psychology constraints on innate vs. learned contributions; (b) public-record and administrative-procedure constraints on claims of government action; (c) electoral-geography constraints on claims of national political capture. For each exemplar record: the prior free parameter, the constraining material, and the resulting reduction in Amb. Use the set only to condition relevance judgment; never as automatic proof.

6. Calibration Notes from Existing Runs

Conscience sketch

- Structuring G from open vector to three named components produced a clear, felt reduction in Amb and was correctly treated as progress without full clearance.

- Default illustrative thresholds ($\tau_{c} \approx 0.95$, $\tau_{a} \approx 0.80$) felt usable; Amb was the binding gate both before and after tightening.

News-article application

- Pre-classifying claims as descriptive vs. normative/strategic would have made Cons and Amb scoring cleaner (advocacy claims systematically carried higher Amb).
- Two search cycles reduced Amb on factual sub-claims (electoral geography, database scale, platform content) while correctly leaving stronger interpretive claims provisional. This calibration supports treating “provisional after serious search” as a stable, honest status rather than a temporary embarrassment.

7. Failure-Mode Logging Structure

Maintain a running log with entries of the form:

```
Date | Domain | Claim or layer summary | Gate outcome | Later evidence
| Direction of error (blocked-but-later-supported / allowed-but-later-collapsed)
| Notes on which rule or judgment contributed | Adjustment made (if any)
```

No failure-mode entries are yet confirmed from the two existing applications; both remain consistent with currently available material. Future applications must populate this log.

8. Lightweight Templates (Ready to Use)

A. Anchor & Claim-Type Template

```
Anchors (L0):
1. ...
2. ...
Claim / Layer element: ...
Pre-classification: [ ] Descriptive [ ] Normative/Strategic [ ] Mixed (split as ...)
```

B. Gate Scoring Sheet

```
Cons: score ____ notes: ...
Agree: score ____ notes: ...
Prod: # productive consequences = ____ list: ...
Amb: free-parameter list + severity weights = sum ____ interpretation: ...
Higher-level review: pass / caution / fail notes: ...
Verdict: Admissible / Provisional / Not admissible
```

C. Gap Extraction & Ranking Sheet

```
Gap description: ...
Impact (0-2): _ Anchor connection (0-2): _ Measurability (0-2): _ Sum: _
Priority order: ...
Search plan / sources to check: ...
```

D. Material Admission Check

```
Candidate material summary: ...
Relevant to targeted gap? Y/N Why: ...
Cons with current anchors + layers? Y/N Why: ...
Admit for incorporation? Y/N
If admitted, effect on Amb / Prod expected: ...
```

9. Status After This Upgrade

The method now has sharper counting rules for Amb and Prod, a required claim-type pre-classification, a scored gap-ranking checklist, initial classical-exemplar guidance, calibration notes from both live runs, a failure-mode log structure, and ready-to-use templates. These changes directly implement the highest-leverage improvements identified. New applications should use the upgraded rules and templates; results should be logged for further calibration and failure-mode detection. Residual judgment remains explicit and is now better scaffolded.

Operational Upgrade — August 2026. Implements measurable gates, claim-type handling, checklists, exemplars, calibration, failure-mode structure, and templates. Use with the Complete Thesis.